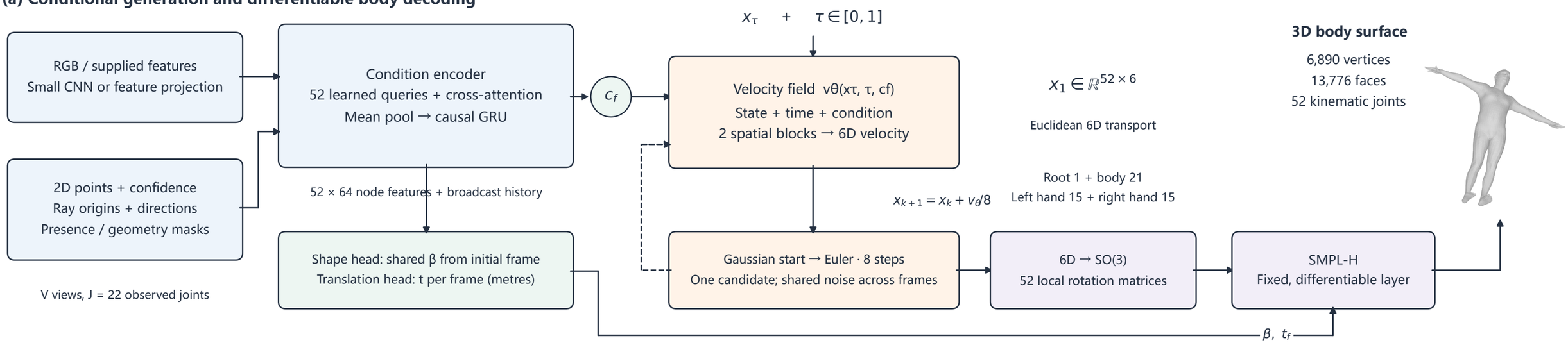
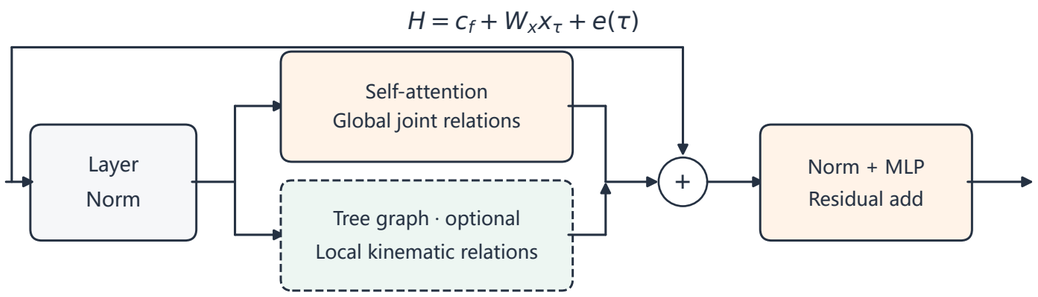
Conditional Flow Matching for SMPL-H Reconstruction

Implemented research baseline | 52 rotation nodes | causal conditioning

(a) Conditional generation and differentiable body decoding



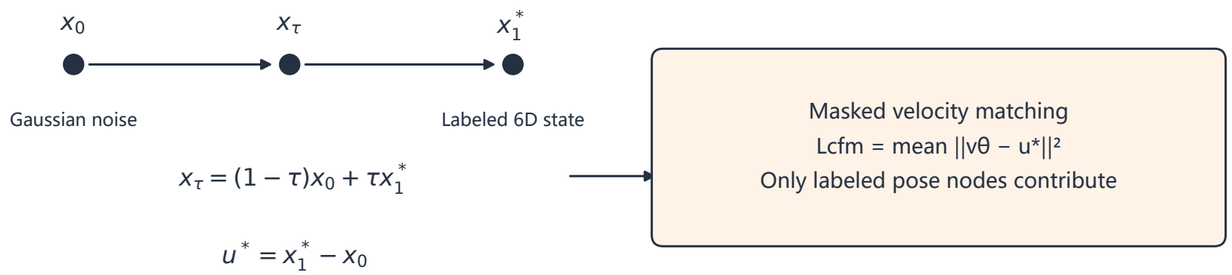
(b) Spatial block • repeated twice



Row-normalized parent–child graph with self-loops; frame-local attention.

Width 64 · 4 heads · output head 64 → 6 per rotation node

(c) Training: linear conditional flow matching



$$L = L_{\text{CFM}} + 0.1L_\beta + L_t + 0.02L_{\text{mesh}}$$

Mesh loss uses a one-step endpoint estimate; full labels required.

Solid arrows: data flow. Dashed loop: sampling feedback. Dashed box: optional graph. f: frame; τ: flow time.

Research baseline; synthetic validation only. Stage, ground and contact remain in the separate engineering pipeline.